

SOFT TISSUE TUMORS

- **Malignant** soft tissue tumors incidence is less than 1% of all cancers; causes 2% of cancer mortality – **usually aggressive**
- **Benign > malignant**; 100 to 1 (except for skeletal muscle neoplasms)
- Most occur in extremities, esp. thigh; 15% arise in children
- Commonly present as a growing mass, sometimes painful, possible loss of range of motion
- Most sporadic; minority associated with mutations in tumor suppressor genes associated with syndromes (ex., Neurofibromatosis 1, Gardner syndrome, Li-Fraumeni, etc)
- Some linked to environment – radiation, burns, toxins
- **Most appear to arise from pluripotent mesenchymal stem cells** – acquire somatic “driver” mutations
- 20% have simple karyotypes - e.g., Synovial sarcoma, Ewing sarcoma
- 80% have complex karyotypes – multiple chromosomal gains and losses, e.g., undifferentiated pleomorphic sarcoma
- Pathological classification combines **morphology, IHC, and molecular diagnostics; grade and stage – important prognostic indicators**

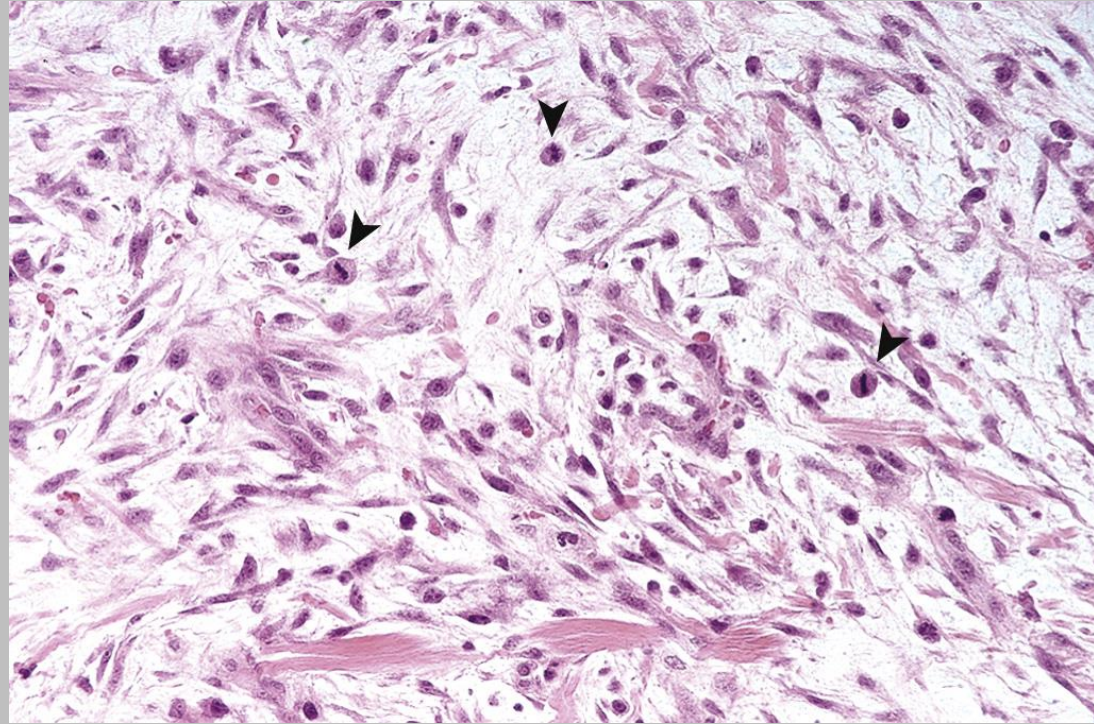
Nodular Fascitis

- Gross: non-encapsulated lesion, well-circumscribed or slightly infiltrative, < 3 cm
- Histo: Plump, immature-appearing **fibroblasts** and myofibroblasts with elongated nuclei showing punctate nucleoli; **frequent mitoses – none abnormal**
- Changes from hypercellular regions with **myxoid** stroma to hypocellular areas with **fibrous** stroma (zonation)
- Storiform or fascicular patterns, common in cellular areas; metaplastic bone, cystic areas, prominent vessels, extravasated red cells, and infiltrating lymphocytes common

Nodular fasciitis

- Plump, randomly oriented spindle cells surrounded by myxoid stroma; “tissue culture” appearance

Note mitotic activity (arrowheads) and extravasated red blood cells



Robbins and Cotran, Pathologic Basis of Disease 10th ed. 2020, Ch.26, Fig. 26.51

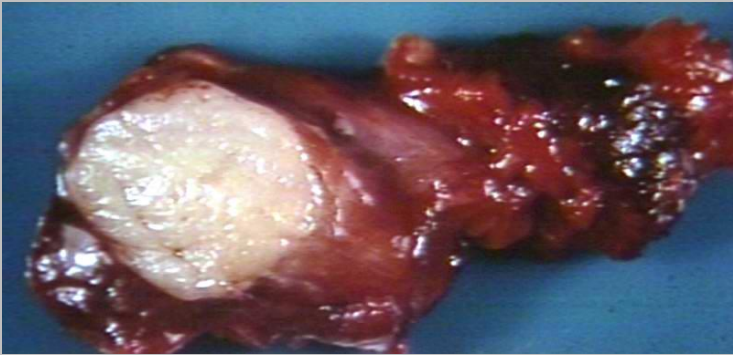
Superficial Fibromatoses

- **Benign growth** - can cause local **deformity, painless, firm, subcutaneous induration**, innocuous clinical course
- M > F males
- Nodular, proliferative lesions made of **plump spindle cells (fibroblasts) arranged in poorly defined long fascicles surrounded by abundant dense collagen**
- Clinical subtypes:
 - **Dupuytren contracture (palmar fibromatosis)**: Irregular or nodular thickening of superficial palmar fascia, unilateral or bilateral; increased incidence with age; attachment to overlying skin: puckering or dimpling, slowly progressive flexion contracture
 - **Ledderhose disease (plantar fibromatosis)**: In boys < 10 years of age into adolescence; unilateral, does not cause contractures (may be associated with palmar and penile fibromatosis)
 - **Peyronie disease: (penile fibromatosis)**: Palpable induration or mass on dorsolateral aspect of penis; may cause abnormal curvature of the shaft and constriction of the urethra
- Palmar and plantar progress in about 50% of cases; some resolve spontaneously; recurrence is common (even after excision)

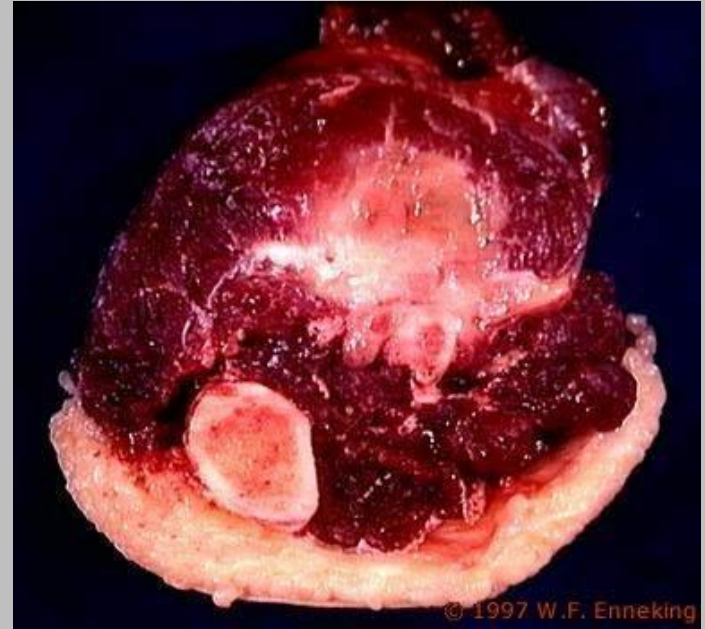
Fibrosarcoma

- **Rare, malignant tumor of fibroblasts** ($\leq 3\%$ of adult sarcomas); most patients 40 - 55 years
- Two types of fibrosarcomas: infantile/congenital-type fibrosarcoma (rarely mets), and adult-type fibrosarcoma
- Many cases formerly called fibrosarcoma where actually dedifferentiated liposarcoma, fibromatosis, fibrosarcomatous DFSP, low-grade fibromyxoid sarcoma, MPNST, synovial sarcoma or MFH-pleomorphic
- Usually **deep soft tissue** of **lower extremities or trunk**, **painful, fixed, soft** tumor, **unexplained weight loss**
- 50% recur, 25% metastasize (lung, bone); more metastases if more cellular and higher mitotic activity
- Survival: 5 year - 41%, 10 year - 29%
- **Diagnosis of exclusion:** IHC and molecular studies identify subtypes
- May appear well circumscribed but non-encapsulated; fleshy, hemorrhagic, necrotic, white-tan
- **Histo: Highly cellular fibroblastic proliferation in herringbone pattern and variable collagen;** scant cytoplasm, tapering elongated dark nuclei with increased granular chromatin, variable nucleoli; **mitotic activity present, often with abnormal forms, stains positive for vimentin, negative for S-100, cytokeratins**
- Treatment : Radical excision, radiation if residual tumor or positive margins; possibly chemotherapy if high grade
- Definitive cause not identified: Variable genetic mutations; predisposing factors-sites of prior injury, chronic inflammation, irradiated tissue, preexisting well-differentiated liposarcoma

Fibrosarcoma



Gross: “fish-flesh” surface



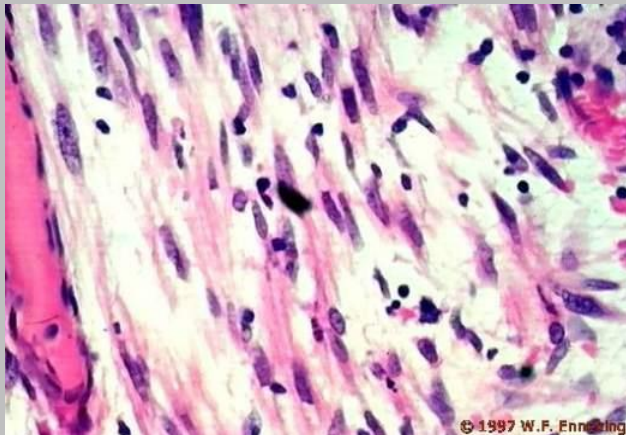
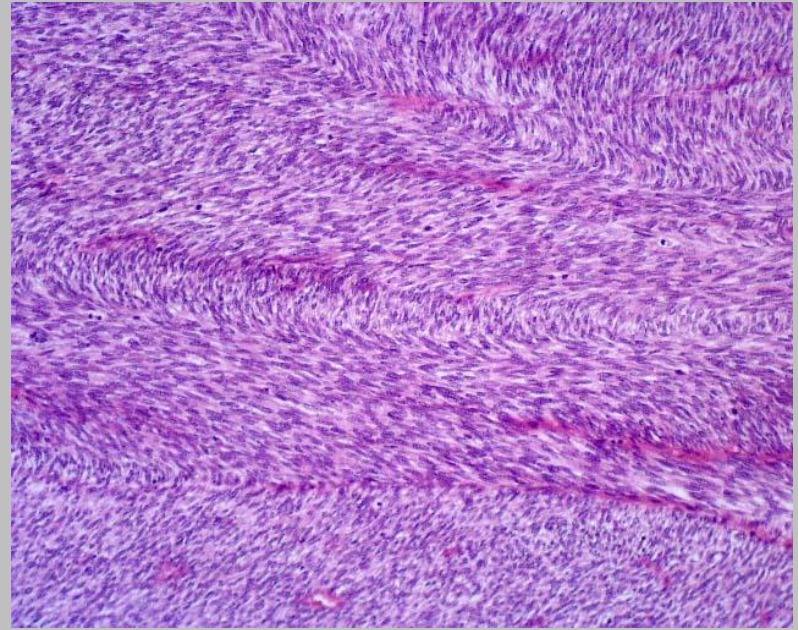
Gross: Infiltrative lesions, “pseudo-encapsulated”, cut section shows white rope-like appearance with lobulation, hemorrhage and necrosis

Fibrosarcoma microscopic

Cellular, spindled hyperchromatic cells with increase nuclear to cytoplasmic ratio; **herring bone pattern (classical)**



“herring bone” pattern



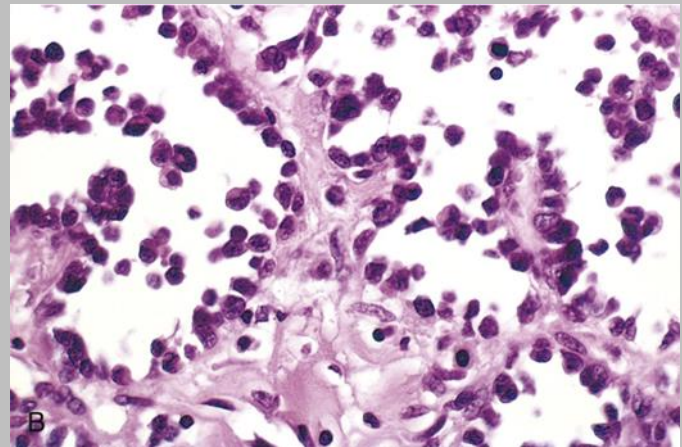
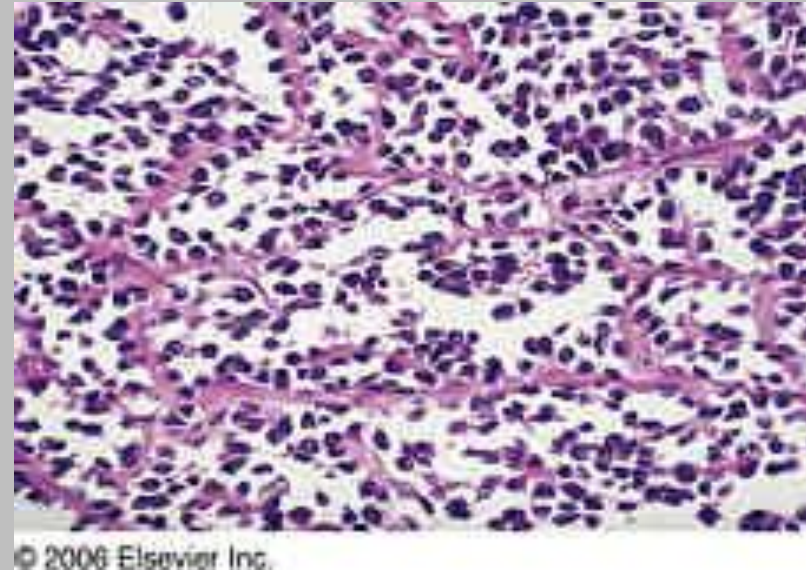
Hyperchromatic nuclei with increased N/C ratio

Rhabdomyosarcoma (RMS)

- **Rare malignant mesenchymal tumor with skeletal muscle differentiation** (But most common soft tissue sarcoma of childhood, cancer in the connective tissue wherein the mesenchymal-derived malignant cells resemble the primitive developing skeletal muscle, usually starts in muscle tissue)
- Four subtypes are recognized: **alveolar (20%), embryonal (50%), pleomorphic (20%)**
- Generally considered to be a disease of **childhood**, as the vast majority of cases occur in those below the age of 18
- Symptoms vary depending on the tumor's location, commonly a **quickly growing mass with pain, unexplained weight loss**
- **Alveolar and embryonal subtypes: Most common soft tissue sarcomas of childhood and adolescence**, usually before 20 years of age (**alveolar**, older children and young adults, **20-35 years-old, arms, legs, or trunk**)
 - **Pediatric rhabdomyosarcomas** often occur in the **sinuses, head and neck, and genitourinary tract**
- **Pleomorphic rhabdomyosarcoma usually occurs in adults** (typically affects adults ages 50 and older); spindle cell/sclerosing type affects all ages, most common locations lower and upper **limbs**
- **Aggressive** and usually treated by **surgical** resection and **chemotherapy**, with or without **radiation** therapy
- Histologic type and location correlated with survival; **sarcoma botryoides (subtype of embryonal rhabdomyosarcoma) has the most favorable prognosis with good results from treatment**; **alveolar** rhabdomyosarcoma four year survival rate without remission for local tumors is 65 percent, while the four year survival rate with **metastatic is only 15%**; **pleomorphic** rhabdomyosarcoma often **fatal**
- Stains positive for **desmin, myogenin**, negative for **S-100, cytokeratins**. Presence of the **PAX-FOXO1 fusion gene** that is found in some patients with RMS is associated with **poorer survival** (due to t(2;13) or t(1;13) translocations, PAX3 is a transcription factor that initiates skeletal muscle differentiation, PAX3-FOXO1 fusion protein interferes with differentiation)

Alveolar Rhabdomyosarcoma

- Micro: Crossed by network of fibrous septae that divide cells into clusters or aggregates, creating crude resemblance to pulmonary alveoli
- Tumor cells: uniform and round with little cytoplasm; cross-striations uncommon

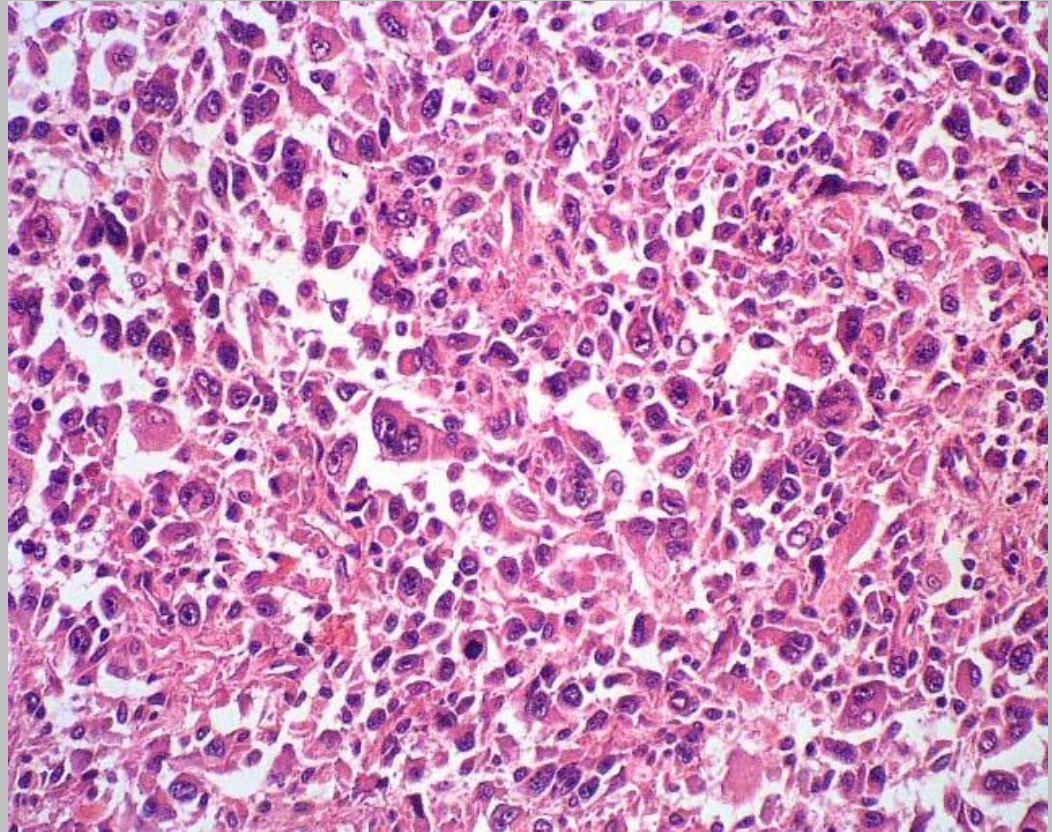


Numerous spaces lined by discohesive, uniform round tumor cells

Robbins and Cotran, Pathologic Basis of Disease, 10 th ed., 2020, Ch.26, Fig. 26.53

Pleomorphic Rhabdomyosarcoma

- Micro: Large, sometimes multinucleated, bizarre eosinophilic tumor cells; can overlap histologically with other pleomorphic sarcomas
- Immunohistochemistry (e.g., myogenin, desmin staining) usually needed to confirm rhabdomyoblastic differentiation



Question

A 6 year old girl has a polypoid mass growing from her vagina which resembles a grape-like cluster of tissue. This mass is a malignant skeletal muscle tumor. What is most likely a feature of this neoplasm?

- A. Microscopic examination reveals areas which crudely resemble lung alveoli
- B. There are not translocations present
- C. Prognosis is better than other malignant skeletal muscle tumors
- D. Immunohistochemistry is needed to confirm fibroblast differentiation
- E. This is the least common soft tissue sarcoma in children

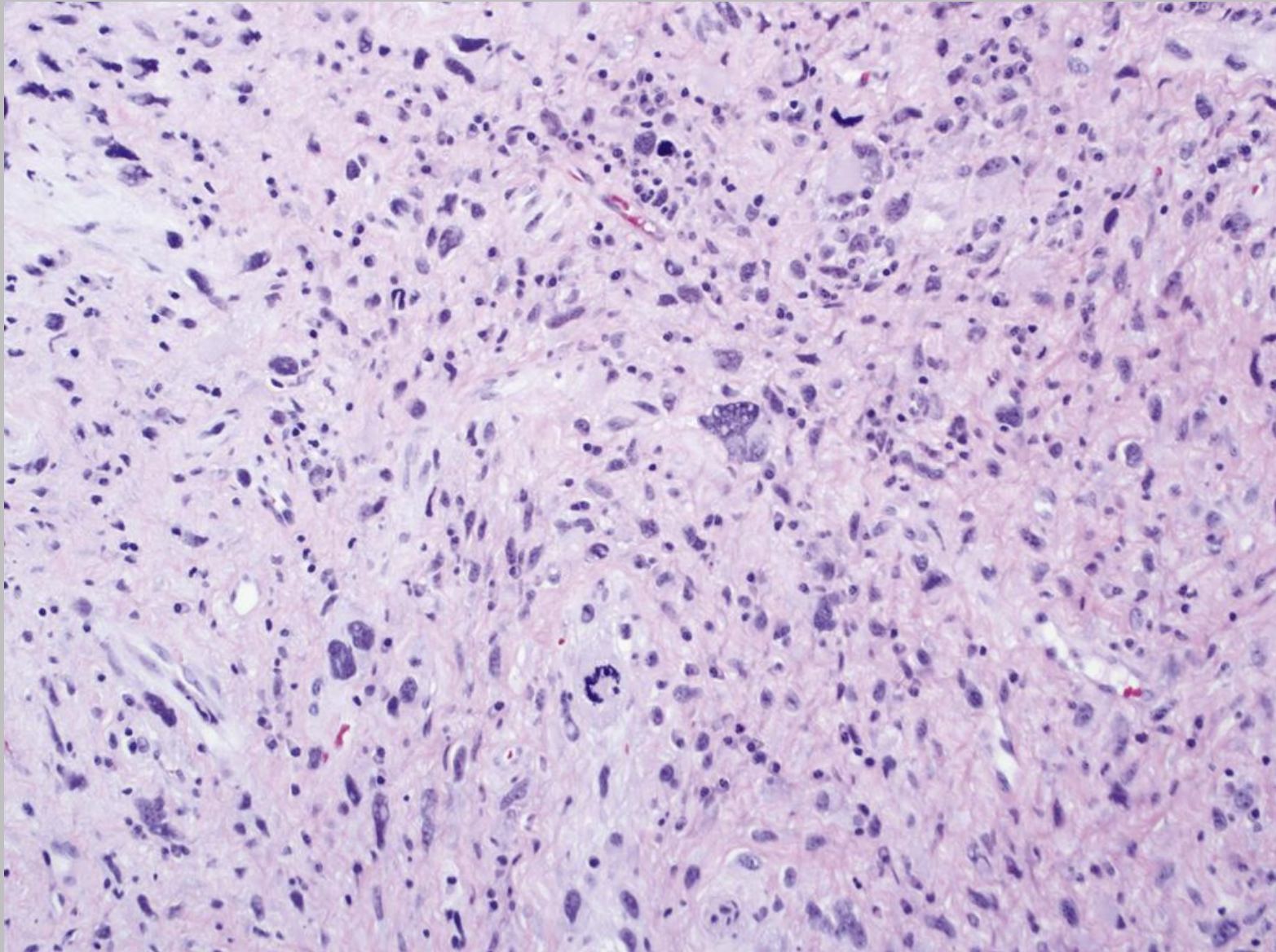
Synovial sarcoma

- The most common location for these tumors is within the **deep soft tissues adjacent to large joints**, e.g. knee and popliteal fossa
- Can present in locations that **lack synovium; uncertain origin**, inconsistent with origin from synoviocytes or synovial structures, e.g. joints, tendon sheaths and bursae
- Approximately 10% of all soft tissue sarcomas, usually **20s to 40s**
- Individuals often present with **deep-seated pain, numbness, limited range of motion**, firm, fixed, lump or **swelling** under the skin, often **around the joints** in the arms or legs
- Symptoms similar to those of arthritis or bursitis, such as swelling and inflammation present for **several years**
- Characteristic chromosomal translocation **t(x;18)(p11;q11)**: Produces fusion of SS18 gene to one of three SSX genes - fusions encode chimeric proteins that interfere with normal chromatin remodeling (the repositioning of nucleosomes on chromatin that influences gene expression)

Synovial Sarcoma

- Micro: Monophasic or biphasic
- Monophasic - uniform **spindle cells** with scant cytoplasm and dense chromatin arranged in short, tightly packed fascicles and **biphasic also contains gland-like structures composed of cuboidal to columnar epithelioid cells**
- **IHC helpful because tumor cells, especially in biphasic type, are positive for epithelial antigens versus most other sarcomas (e.g., cytokeratins, epithelial component is consistently positive for CK7, CK8, CK14, CK18, CK19; spindle cell component usually focal and variably positive for CK7, CK19, CK18, CK8, CK14, CK17, CK20)**
- Treated aggressively with limb-sparing surgery and chemotherapy
- 5-year survival varies from 25% to 62%, related to tumor stage and patient age
- In general, synovial sarcoma has a survival rate of 50-60% at 5 years and 40-50% at 10 years
- The survival rate for synovial sarcoma is lower in cases where the cancer has returned or spread
- Treatment involves surgery, chemotherapy and radiation. Some factors for prognosis include the following: Small tumor size—less than 5cm wide, few mitoses, easy to remove by surgery
- Common sites of metastases are the lung and, **unusually for sarcomas, regional lymph nodes**

Undifferentiated pleomorphic sarcoma showing anaplastic spindled to polygonal cells



Summary Slide

- Lipoma – benign, adipose tissue
- Liposarcoma – malignant, adipose tissue, subtypes well-differentiated, myxoid, pleomorphic
- Nodular fasciitis – benign, fibroblasts
- Superficial dermatoses – palmar, plantar, penile
- Deep fibromatosis – recur, associated with APC and beta-catenin gene versus superficial fibromatosis (both fibroblasts/myofibroblasts)
- Fibrosarcoma – malignant, aggressive, fibroblasts, recur, metastasize, herring bone histological pattern
- Rhabdomyosarcoma – malignant, skeletal muscle
 - Embryonal – subtype Sarcoma Botyoides (best prognosis)
 - Alveolar – t(2;13)
 - Pleomorphic
- Synovial sarcoma – malignant, t(X:18), monophasic or biphasic
- Undifferentiated pleomorphic sarcoma – highly malignant, need special stains and studies,